(n = 1) spine and the number of involved vertebrae was 1 (n = 2) and 2 (n = 2). The epidural extension ranged between Bilsky score 0, 1a, 1b, and 2, and paraspinal involvement was observed in 3 of 4 cases. A step-wise planning procedure was performed such that all stages of planning from initial CT-MR image registration to gross-tumor volume (GTV) delineation to clinical target volume delineation (CTV) to planning target volume (PTV) definition, and treatment planning were each analyzed independently. This was achieved by providing consensus results to the group after each step of the planning, forming the basis for the next step.

Results: Co-registration of CT and MR images was performed using automatic registration in all centers, but all centers agreed that the results required manual adjustment. Averaged over the four cases, registration variability ranged between 0.8 mm (anterior-posterior) and 1.3 mm (superior-inferior) and 0.9° (around anterior-posterior axis) and 2.0° (around left-right axis) with one standard deviation (SD). Clinically significant inter-observer variability was observed for GTV delineation, which was 8.8 mm (1 SD) averaged over all cases and six directions; variability ranged between 3.4 mm in the inferior direction and 11 mm in the superior direction. This variability in GTV delineation is mostly explained by disagreement about disease extension within the vertebral body in two cases. Based on the consensus GTV, there was less variability in the process of CTV delineation, which was 2.1 mm (1 SD) averaged over all cases and six directions and ranged between 1.6 mm in inferior and 4.2 mm in the left direction. Applied CTV-to-PTV margins ranged between 0 mm and 3 mm. A detailed demonstration of the differences in the target volume concepts, the delineation variability and the treatment planning and dose distributions was produced.

Conclusions: Amongst a group of experienced spine RS centers, clinically significant variability in all steps of the treatment planning process was demonstrated. This may translate into differences in patient clinical outcome and highlights the need for consensus criteria.

Author Disclosure: M. Guckenberger: E. Research Grant; This research was partially supported through an Elekta research grant with all institutions being members of the Elekta Spine Radiosurgery Research Consortium. This work and these data, however, are. R.A. Sweeney: None. J.C. Flickinger: None. I. Grills: None. N. Tyagi: None. A. Sahgal: None. D. Letourneau: None. J. Sheehan: None. D. Schlesinger: None. P.C. Gerszten: None.

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Correlation of Local Failure With Measures of Dose Insufficiency in the High-dose Single-fraction Treatment of Bony Metastases

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Purpose/Objective(s): The probability of local control of metastatic spinal and paraspinal lesions using high dose single fraction image-guided radiation therapy has been shown to be dependent on the prescription dose, with a dose response between 18 Gy and 24 Gy reported. The close proximity of the spinal cord frequently leads to the existence of cold spots in the target coverage, however. In this analysis, we investigate the significance of these cold spots by examining the correlation of dose insufficiency for the gross tumor volume (GTV) with local treatment failure.

Materials/Methods: Between July 2003 and January 2011, 329 patients with a total of 413 spinal and paraspinal metastatic tumors were treated with a high-dose single-fraction radiation therapy. The prescribed dose was 18 - 24 Gy, and the maximum spinal cord / cauda equina dose was 12 - 14 Gy. We investigate the correlation of local control with several measures of dose insufficiency of the gross target volume (GTV). The minimum doses delivered to the hottest 95%, 98%, and 100% (D95, D98, and Dmin) and the percent volume receiving at least 95% of the prescription dose (V95) was computed. The distributions of these measures for the treatments with local failure were compared with the patient group taken as a whole using a Wilcoxon dose sum statistic.

Results: Local failure has been identified radiographically in 18 patients. Of the treatments that resulted in local control and that could be retrieved from archive, 187 have been analyzed to date in this preliminary analysis. The mean Dmin D98, D95 and V95 for local failure were 13.7 Gy, 18.3 Gy, 20.0 Gy, and 89.6%. For the group with local control, the corresponding values are 16.5 Gy, 20.2 Gy, 21.9 Gy, and 94.6%. The distributions of Dmin, D98, D95, and V95 for the GTV were all found to be significantly different from the distributions of all patients analyzed to date. The corresponding p values are 0.007, 0.016, 0.008, and 0.020.

Conclusions: The results indicate that measures of tumor dose insufficiency such as D95, D98, Dmin, and V95 may be important risk factors for local failure. Although two failures occurred with Dmin >15 Gy (18.7 Gy and 19.5 Gy), the risk of local failure for patient treatments where Dmin = 15 Gy, suggesting that this metric may be an important predictor of local control.

Author Disclosure: D. Lovelock: None. M. Tam: None. A. Jackson: E. Research Grant; NCI RO1CA129182. M.H. Bilsky: None. E. Lis: None. B.W. Cox: None. Y. Yamada: G. Consultant; Varian Medical Systems, Continum Medical Institute.

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I-125 Episcleral Plaque Brachytherapy for Uveal Melanoma: A 15-year Single Institution Experience

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Purpose/Objective(s): To determine the long-term outcomes of I-125 episcleral plaque brachytherapy for uveal melanoma and identify dosimetric factors associated with ocular complications and tumor control.

Materials/Methods: Medical records were reviewed for 494 patients treated with I-125 plaque brachytherapy for uveal melanoma from 1996 to 2011. Brachytherapy records for 436 patients were reviewed to analyze radiation doses at specific dose reference points. Ophthalmologic follow-up records were analyzed to assess the incidence of recurrence and ocular complications after therapy. Survival curves were estimated using the Kaplan-Meier method. Student's t test and Fisher's exact test were used to analyze the relationships of dose delivered and type of plaque to outcomes.

Results: Mean tumor thickness and long basal diameter were 4.9 mm and 12.1 mm, respectively. The median plaque size was 18 mm. One hundred ten plaques (22.6%) required a notch due to proximity to the optic nerve. The median follow-up was 42 months (range, 6-175 months). The estimated 5-year overall and cause-specific survivals were 73.2% and 88.6%, respectively. Local control was 95.3% with 26 local failures, and 56 patients (11.3%) developed metastatic disease. The mean doses to the tumor apex, tumor base, and prescription point were 98.8 Gy, 269.4 Gy and 85.6 Gy, respectively. The mean dose to the tumor base was similar in those with local control compared to those with local failures (269.1 Gy vs. 261.7 Gy; $p = 0.74$). The corresponding radiation dose to the tumor apex was 99.1 Gy versus 94.2 Gy ($p = 0.15$). Thirty-eight patients (6.8%) underwent enucleation. The reasons for enucleation included local recurrence (20 patients), second primary tumor (1 patient) and severe radiation toxicity (17 patients; 3% of treated patients). The mean dose to the tumor base in patients requiring enucleation was significantly greater than in those without enucleation (323.3 Gy vs. 263.5 Gy, $p = 0.02$). The corresponding mean radiation dose to the tumor apex was 95.1 Gy and 99.2 Gy ($p = 0.11$). Patients who required enucleation secondary to radiation toxicity had a significantly greater dose to the tumor base than those without enucleation (392.1 Gy vs. 263.7 Gy, $p = 0.001$). The corresponding mean radiation doses to the tumor apex were 92.3 Gy and 99.2 Gy ($p = 0.07$). There was no significant association between presence of a notch or size of the plaque and recurrence or enucleation (all $p > 0.05$).

Conclusions: I-125 episcleral plaque brachytherapy leads to excellent local control and globe preservation. There is no association between need for a notched plaque or plaque size with tumor recurrence or need for enucleation. Patients requiring enucleation had higher a radiation dose to the base of the tumor.

Author Disclosure: S.N. Badiyan: None. R.C. Rao: None. A.J. Apicelli: None. S. Acharya: None. A.A. Garsa: None. T. DeWees: None. J. Garcia-Ramirez: None. P.W. Grigsby: None. J.W. Harbour: None.